Linguistic Differences Between Phishing and Legitimate Emails

A feature-based NLP and text analytics study | Concise project report | Original term paper: 2026

Research purpose

This project investigates whether linguistic patterns in phishing and legitimate email bodies differ systematically. It measures how senders address recipients, express urgency or uncertainty, use emotion-related vocabulary, and structure their writing. The original research uses Interpersonal Deception Theory as a framework for interpreting these patterns; it does not train a phishing-detection classifier.

Scope at a glance

Initial corpus	Final English sample	Feature groups
2,000 human-authored emails (1,000 / class)	1,148 emails: 695 legitimate + 453 phishing	Pronouns, modality, stance, urgency, emotion, style

Five-step research workflow

- 1. Acquire: select human-authored legitimate and phishing subsets from the Kaggle dataset identified in the thesis; exclude generated-message files.
- 2. Clean: normalize text for duplicate checks; reduce 2,000 messages to 1,193 unique texts; exclude 36 non-English messages and nine messages shorter than 20 English alphabetic words, while manually reviewing flagged cases.
- 3. Extract: compute 19 linguistic and stylistic indicators. Normalize word counts per 100 alphabetic words; compute MATTR-20 and Flesch Reading Ease on appropriate text representations.
- 4. Test: compare group means using Welch independent-samples t-tests; apply Holm correction across 19 comparisons and report signed Cohen's d.
- 5. Check robustness: repeat the analysis on a balanced 453 phishing / 453 legitimate subset (reported random seed: 42); interpret results through communication theory.

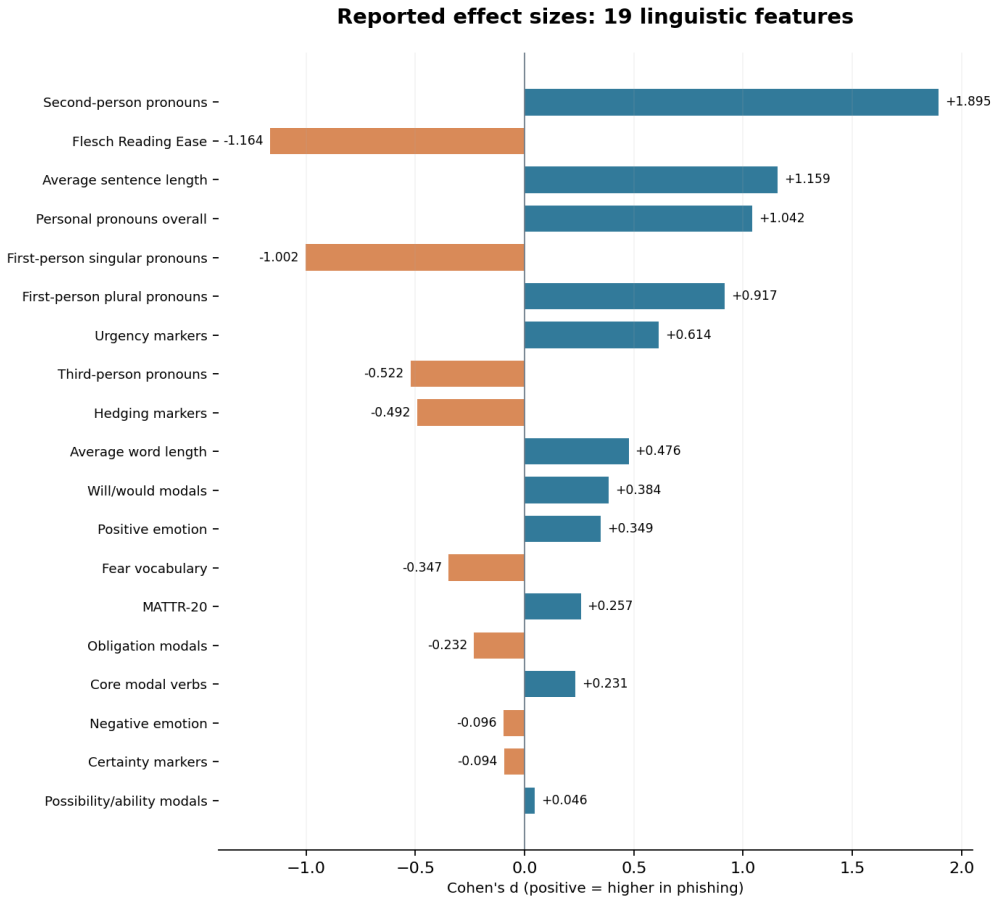
Provenance and scientific status

The 48-page original term paper is titled Linguistic Differences Between Phishing and Legitimate Emails: A Feature-Based NLP and Text Analytics Study and names Adil Hussain as its author (HSE University, 2026). The findings summarized here are reported in that document. The original message-level data, preprocessing scripts and NRC lexicon were not included in the supplied project archive, so the numerical results could not be rerun independently.

Findings from the original term paper

The thesis reports 16 of 19 linguistic features statistically different between groups after Holm adjustment; three were not significant: certainty markers, negative-emotion vocabulary, and possibility/ability modals.

Feature	Legitimate mean	Phishing mean	Reported Cohen’s d
Second-person pronouns (per 100 words)	1.191	5.726	+1.895
Urgency markers (per 100 words)	0.026	0.448	+0.614
Average sentence length (words)	9.196	19.524	+1.159
Flesch Reading Ease	67.847	48.993	−1.164
First-person singular pronouns	1.982	0.211	−1.002



Source: original term paper, Table 2, pp. 21–22. Not independently reproduced.

Effect sizes have been transcribed from thesis Table 2 (pp. 21–22) and visualized; they are not independent model outputs. Positive effect sizes mean a higher group mean for phishing.

Interpretation, implementation and limits

How to interpret the findings

In this particular corpus, phishing messages used more second-person forms (such as you and your), more urgency markers and longer sentences; legitimate emails were more readable on average and used more first-person singular references. The results do not imply that a single word or style identifies deception. Group differences may also reflect source composition and email genres.

What the GitHub repository contains

- A professional README, project provenance record, data/ethics notes and the historical 19-feature summary table.
- A newly written Python extractor that demonstrates pronouns, modal groups, hedging, urgency and style features; optional locally licensed NRC emotion features.
- Independent statistical utilities (Welch test, Holm adjustment and Cohen's d), original-data effect-size visualization and fictional demo inputs.
- Ten passing source-code unit tests plus command-line instructions and this report. These checks verify the new companion software, not the historical thesis estimates.

How to run the demonstration

From the repository root, install **requirements.txt**, set **PYTHONPATH=src**, and run **python -m phishing_language.cli demo**. This generates feature tables and illustrative group comparisons from eight fabricated messages. For authorized email-level data, use the **extract** and **compare** commands documented in README.

Reproducibility, privacy and publication constraints

- The supplied source materials do not include the historical raw emails, analysis scripts, complete dependency versions, or exact text preprocessing settings. The original 16/19 result therefore remains a documented claim, not a reproduced finding.
- The illustrative feature pipeline uses approximate sentence segmentation and syllable counting; it should not be represented as an exact reimplementation of the thesis.
- Original email bodies and NRC Emotion Lexicon data are excluded from the public package. Obtain and review their licenses independently; remove personal information and any sensitive email excerpts before distribution.
- The full 48-page thesis PDF is not bundled publicly until the owner confirms authorship, rights and any institutional assessment conditions.

Conclusion and portfolio value

This is a defensible research showcase in computational linguistics and interpretable text analytics. It demonstrates feature design, corpus preprocessing, multiple-comparison control, effect-size interpretation and careful reasoning about deceptive communication. The repository does not claim phishing prediction accuracy, operational detection capability or peer-reviewed publication.

Reference: Adil Hussain (2026), Linguistic Differences Between Phishing and Legitimate Emails: A Feature-Based NLP and Text Analytics Study, HSE University master's term paper. Details and limitations: README and docs/METHODOLOGY_AND_LIMITATIONS.md.